

Exercise Sheet 4

Deadline for submission: Wednesday, 22 Oct 2025, 08:00.

Discussed in exercise session: Friday, 24 Oct 2025, 10:15.

Exercise 4.1 - Structure of a Singular Value Decomposition (SVD)

(Submit your solutions in the notebook.) Let a non-square matrix be given by $A \in \mathbb{R}^{m \times n}$ (with $m > n$ and $\text{rank}(B) = r$).

1. Consider its full SVD given by $A = USV^t$.
Specify the dimensions for U , S , and V .
2. What is the difference to the economy version of the SVD?
Specify the dimensions of the resulting U_* , S_* , V_* for the economy version, with $A = U_* S_* V_*^t$.
3. Use the SVD for the matrix A to show that the Eigenvalues of $A^t A$ are contained in $S^t S$, i.e. $(A^t A)V = V(S^t S)$.
Hint: Remember the properties of the matrices U and V with respect to their inverses.
4. What do the columns of U contain? What do the columns of V contain?
Hint: The results of (3.) help to remember this.

Exercise 4.2 - Under-/Overdetermined Linear Systems

We are given the following linear systems. Are they well-defined, underdetermined or overdetermined? Provide an explanation.

1.

$$A_1 \mathbf{x}_1 = \mathbf{b}_1, \quad \text{with} \quad A_1 = \begin{pmatrix} 5 & 0 & -1 \\ -2 & 2 & 3 \end{pmatrix}, \quad \mathbf{b}_1 = \begin{pmatrix} 3 \\ 2 \end{pmatrix},$$

2.

$$A_2 \mathbf{x}_2 = \mathbf{b}_2, \quad \text{with} \quad A_2 = \begin{pmatrix} 5 & 0 & -1 \\ -2 & 2 & 3 \\ 3 & 2 & 2 \end{pmatrix}, \quad \mathbf{b}_2 = \begin{pmatrix} 3 \\ 2 \\ 5 \end{pmatrix},$$

3.

$$A_3 \mathbf{x}_3 = \mathbf{b}_3, \quad \text{with} \quad A_3 = \begin{pmatrix} 5 & 0 \\ -2 & 2 \\ 3 & 2 \end{pmatrix}, \quad \mathbf{b}_3 = \begin{pmatrix} 3 \\ 2 \\ 4.5 \end{pmatrix}.$$

We can solve the systems using singular value decomposition, i.e. with

$$A_i = U_i S_i V_i^t \quad \forall i \in \{1, 2, 3\}$$

Solve the Under-/Overdetermined Linear Systems using this decomposition in Python - a template is provided in `svd_empty.ipynb`. What can you tell us about the solution you get?

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Exercise 4.3 - Zeroing

Let matrix A be of size 4×4 with elements

$$A_{ij} = \frac{1}{100 \cdot i + 0.01 \cdot j}.$$

Also, assume a 4×1 vector $\mathbf{b}$ with

$$b_i = \sum_{j=1}^4 A_{ij}.$$

Write a Python script in order to do the following:

1. Compute A , its SVD and its condition number.
Zero the smallest singular value and calculate the resulting $B = U\bar{S}V^t$.
2. Compute the ranks of A and B .
3. Solve $A\mathbf{x}_1 = \mathbf{b}$ and $B\mathbf{x}_2 = \mathbf{b}$ using the above SVD.
4. Why is it necessary to set the smallest singular value to zero?

A template is provided in `svd.empty.ipynb`.